

STC 412

(1)

PHYSICAL CHEMISTRY (Thermodynamics)

INTRODUCTION

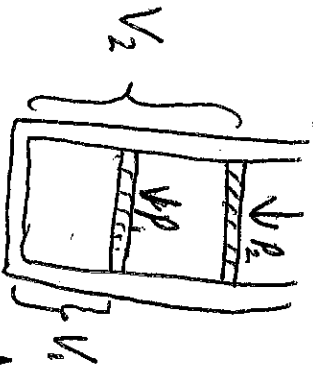
One of the most fundamental manifestations in nature is energy that accompanies all changes and transformation. This can be, by absorption, emission or by the redistribution of energy. An integral property of matter is motion which as matter itself cannot be destroyed. The motion of matter has different forms that can be transformed one into another, and the measure of the motion of matter is energy. This energy can be expressed quantitatively in a different ways through those parameters characterizes of each specific form of motion and a unit specific for this form.

The object of thermodynamics is to seek out the relationship between kinds of energy and their diverse manifestations. The laws of thermodynamics govern the transformation of one kind of energy into another and are based on experimental facts. The term thermodynamics literally implies a field that deals with the mechanical action produced by heat.

Thermodynamics is a branch of physical sciences that treats various phenomenon of energy and related properties of matter, especially in the transformation of heat into other forms of energy and vice versa.

Examples

(1) Heat to work process; $E = W = P \Delta V$.



$$\Delta V = V_2 - V_1$$

$E = \text{Energy}$
 $W = \text{Work done}$

$P \Delta V = \text{Pressure that cause a unit change in } V$.

- (2) Electrically to heat
(3) Work to kinetic energy i.e.;

$$W = \frac{1}{2} m v^2$$

This transformation of energy can be collectively be referred to in the law of conservation of matter or energy, which states that 'energy cannot be created nor destroyed'.

DEFINITION OF SYSTEM:

System - Any collection of atoms, molecules or ions set in an environment may be regarded as constituting a system with definite boundaries separating it from separate world. It is that part of the universe whose properties of interest are under investigation.

A system may be homogeneous if every parameter of the system has the same value in all part of the system or continuously changing from point to point. A system is said to be heterogeneous if it consists of several macroscopic part separated from one another by visible interfaces. Some parameters changed in jumps at these interfaces.

② The homogeneous part of a system separates from other parts by visible interface are called **Phases**. Each phase of a system is characterized by its own equation of state by the type, $f(P, V, T) = 0$.

This equation of state relates the pressure, Volume, Temperature and other parameters of a system (phase).

Thermodynamics is interested mainly in the initial and final state of a system without any anxiety about the speed of the particle reacting. It chiefly deals with such state of a system when a number of its properties—temperature, pressure, electrostatic potential etc. do not spontaneously change with time and have the same values at all time within the volume of the separate phases. Such state are called equilibrium.

And with an uneven distribution of temperature, pressure, composition and so on inside the phases that also varies with time, the system is known as non-equilibrium.

SURROUNDINGS: The regions outside the boundaries of the system are termed surroundings. However, the surroundings of the system are restricted to a region in the immediate vicinity of the system under study. The boundaries which separate the system from its surroundings are experimentally well defined.

UNIVERSE: A system and its surroundings. ~~Consider~~ Consider the Universe.

Classification of systems:

- (a) Open system: An open system is one which can exchange both energy and matter with its surroundings.
- (b) Closed system: This allows only exchange in energy with surroundings, no matter is involved.
- (c) Isolated system: There is no exchange of matter, no energy is possible with the surroundings.

EXTENSIVE PROPERTIES: - The magnitude of an extensive property depends on the amount of substance present. e.g. total mass, Volume and energy.

INTENSIVE PROPERTY: - This is independent on the total amount but depends instead on the concentration of the substance in the system. This intensive properties are pressure, density, chemical potential (mass, Volume or energy per mole).

HEAT AND WORK.

The form of transformation of motion itself and the related conversion of energy can be divided into two (2) groups viz:-

— The first one include only one form of the transfer of motion by the chaotic collisions of molecules of two bodies in contact i.e. by heat conduction. This measure of motion is what is called heat.

- ③ - The second group includes different form of motion whose general feature is the motion of microscopic bodies containing very great number of molecules under the action of forces. Examples -
1. - Lifting a body against gravitational energy (work)
 2. - transmission of electricity from high to lower electrostatic potentials.
 3. - Expansion of gas under pressure.

This measure of ~~heat~~ ~~pressure~~ energy & work. Heat and work characterized two (2) quantitatively and qualitatively different forms of transmitting motion from a given part of the material wall to another.

First Law of Thermodynamics

This is merely the law of conservation of matter & energy. The chemical energy of the substance is expressed as its internal energy E . If Q = quantity of heat added to a system goes to increase the internal energy and also to do any outside work the system may perform i.e. within the system.

If the increase in energy is ΔE ,

$$E = \Delta E$$

$$\text{or } E_2 - E_1 = \Delta E$$

W = work done by the system on the surrounding

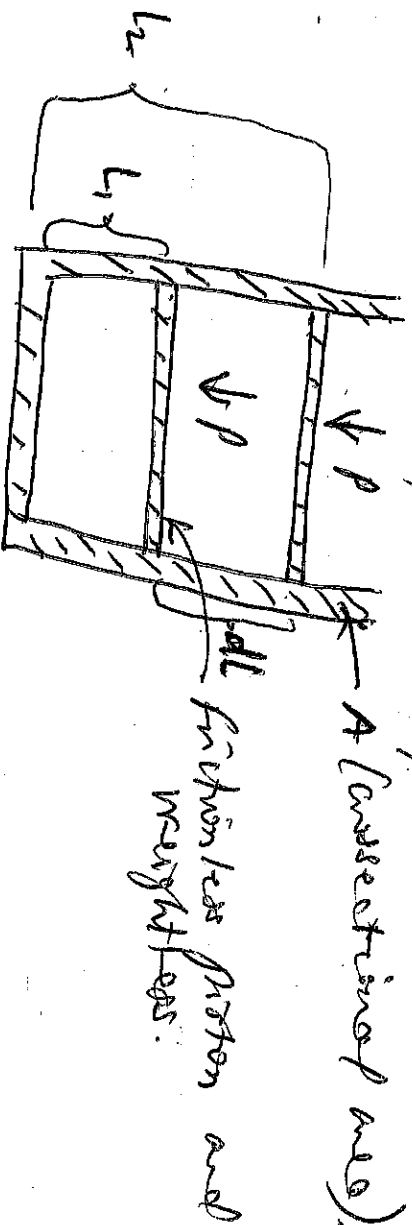
$$\boxed{Q = \Delta E + W} \quad \text{--- (1)}$$

$$\text{or } \Delta E = Q - W.$$

But E depends on the state of the system and not on the manner the changes accomplished.

$$\delta E = E_2 - E_1, \text{ --- (2)}$$

Considering a cylinder of cross section of area, A .



$$l_2 - l_1 = dl$$

$$\text{but } P = f/A$$

$$\therefore f = PA$$

moving the piston dl ;

$$dw = f dl = PA dl \text{ --- (3)}$$

$$\text{but Area, } A dl = dV$$

$$\therefore \int dw = \int P dV$$

$$W = P \int dV \text{ --- (4)}$$

Substituting (4) into (1)

$$q = \delta E + P \int dV \text{ --- (5)}$$

$$\sim \delta E = q - P \int dV$$

Both equations (4) and (5) are perfectly general and applied to equation of q , W and δE in any expansion and contraction system.

< this -

(4)

Volume = constant

$$\therefore V_1 = V_2 \Rightarrow V_1 - V_2 = 0$$

$$\therefore \Delta V = 0.$$

$\therefore$ equation (4) becomes

$$W = P \int_1^2 dV = 0$$

$$\Rightarrow W = 0.$$

equation (1) becomes -

$$q = \Delta E + W$$

but if $W = 0$

$$\therefore q = \Delta E$$

$$q_V = \Delta E$$

(ii) Pressure = 0

$$W = 0$$

$$q = \Delta E$$

(iii) $P = \text{constant}$

$$W = P(V_2 - V_1)$$

Substitute equation (2) into (5).

$$\Delta E = q - P(V_2 - V_1)$$

$$q_P = \Delta E + P(V_2 - V_1)$$

$$q_P = E_2 - E_1 + P V_2 - P V_1$$

$$q_P = (E_2 + P V_2) - (E_1 + P V_1) \quad \dots \quad (5a)$$

Since P and V are thermodynamic properties of the system, it follows that $(E + PV)$ just like internal energy, E are thermodynamic properties of the system. This extensive properties of state

$(E + PV)$ is called the heat content or En from the Great Work, enthalpy always warm.

ie $H = E + PV$ ————— (9)

$$H_2 - H_1 = \Delta H = \Delta E + P \Delta V = \Delta E + \delta P \Delta V \text{ ————— (9a)}$$

At constant P,

$$\Delta H = \Delta E + P \Delta V \text{ ————— (10)}$$

Compare equations (8b) and (10).

$$E_2 + PV_2 = H_2$$

$$E_1 + PV_1 = H_1$$

$$\therefore g_P = H_2 - H_1 = \Delta H$$

$$g_P = \Delta H \text{ ————— (11)}$$

Comparing equation (6a) and (8a) is at constant

$$g_{PV} = \Delta E$$

$$g_P = g_V + P(V_2 - V_1)$$

$$= g_V + P \Delta V$$

$$\therefore g_P = g_V + P \Delta V \text{ ————— (12)}$$

Also, if $V_2 > V_1$, then $W = +ve$, expansion (work on the

$V_2 < V_1$, then $W = -ve$, contraction (environment doing work to the system)

NB: H is a property of state, depend on $V, P, \Delta T$ of the system and possibly the no of mole (n) of the system. A change of $f(P, V, T, n)$ will also produce a change on the system.

If $\Delta H = -ve$ ——— Exothermic (bond in along)
 $\Delta H = +ve$ ——— Endothermic (in bond breaking)

REVERSIBILITY CHANGES (Maximum work)

(5)

Any process which is to be conducted that at any given state, the driving force is only infinitesimally greater than the opposing force and which can be reversed by increasing the reversing force that kind of work is called a reversing process. Otherwise, the process is irreversible.

Maximum work is obtained from a system when any change takes place in it is entirely reversible. To achieve reversible expansion, we set P_{ex} (external pressure equal to P (pressure) at each stage of the expansion. In practice, this qualification could be achieved by gradually removing weights from the piston so that the downward force due to the weights always matches the changing upward force due to the pressure of the gas. When we set $P_{\text{ex}} \geq P$, then work done is:

$$dw = -P_{\text{ex}} dv = -P dv \quad \text{---} \quad (13)$$

The total work done by the reversible expansion is therefore

$$W = - \int_{v_1}^{v_2} P dv \quad \text{---} \quad (14)$$

The above equation (14) can be evaluated since we know how the pressure of the confined gas depends on its volume.

ISOTHERMAL REVERSIBLE EXPANSION

In an isothermal process, the temperature is maintained constant during the entire operation. Since the internal energy of an ideal gas varies with temperature only, imposition of constant temperature implies constancy of E i.e. for an isothermal process,

DEED

Hence, the equation $\Delta E = q - W$ reduces to $\boxed{q = W}$ --- (2)

from the earlier discussion, the total work obtained in the expansion of a gas from initial volume V_1 to the final volume V_2 is equal to the integral of the pressure times the differential of the volume i.e

$$Q = W = \int_{V_1}^{V_2} P dV$$

Thus, in an isothermal process, all the work performed is obtained at the expense of absorbed heat and vice-versa. The magnitude of Q will thus vary with W and may range from Zero for a free expansion to a maximum for reversible process.

For the maximum work in an isothermal reversible expansion of an ideal gas, an expression can be obtained as follows:-

Since, $W_{\text{max}} = \int_{v_1}^{v_2} p \, dv$ — — — (1)

but $PV = nRT$ for an ideal gas,
so $P = \frac{nRT}{V}$

∴ Equation (4) becomes:-

(6)

$$W_{\max} = \int_{V_1}^{V_2} \frac{nRT}{V} dV \quad \text{--- (5)}$$

$$= nRT \int_{V_1}^{V_2} \frac{dV}{V}$$
$$= nRT \ln \frac{V_2}{V_1}$$

$$\text{or } W_{\max} = 2.303 nRT \log \frac{V_2}{V_1} \quad \text{--- (6)}$$

Equation (6) gives the maximum amount of work obtainable in the isothermal reversible expansion of n moles of an ideal gas from Volume V_1 to V_2 at constant temperature T . Again since the temperature is constant,

$$\frac{V_2}{V_1} = \frac{P_1}{P_2}$$

Therefore Equation (6) can be rewritten as

$$W_{\max} = nRT \ln \frac{P_1}{P_2}$$

$$\text{or } W_{\max} = 2.303 nRT \log \frac{P_1}{P_2} \quad \text{--- (7)}$$

For real gas, $\Delta E = 0$ under isothermal conditions because, when temperature is constant, internal energy will change due to change in Volume or Pressure. The same is true for ΔH .

Example 1.

Calculate the max. work obtainable by the bestmost reversible expansion of 2 moles of nitrogen from 10 litre to 20 litres at 25°C.

Solution:

$$\text{from } W_{\max} = nRT \ln \frac{V_2}{V_1}$$

$$\text{but } n = 2 \text{ moles, } \frac{V_2}{V_1} = 20 \text{ litres, } V_1 = 10 \text{ litres, } R = 1.987 \text{ cal deg}^{-1} \text{ mole}^{-1}$$

$$\text{and } T = 273.2 + 25 = 298.2 \text{ K.}$$

$$\therefore W_{\max} = 2 \cdot 303 \times 2 \times 1.987 \times 298.2 \times \log \frac{20}{10} \\ = \underline{\underline{822 \text{ calories}}}$$

Example 2

An ideal gas undergoes a reversible isothermal expansion from an initial volume V_1 to a final volume V_2 and thereby does 10,000 cal. of work. The initial pressure was 10 atm. (a) Cal. V_1 . (b) If the temperature was 2 moles of gas, what must its temperature have been?

Solution

$$(a) \text{ since } W = 2 \cdot 303 nRT \log \frac{V_2}{V_1} \\ = 2 \cdot 303 nRT \log \frac{10}{V_1} \\ \approx 2 \cdot 303 nRT$$

$$\text{but } P_1 V_1 = nRT = W$$

$$\approx 5 \text{ } 1000 \times V_1 = \frac{10,000}{2 \cdot 303}$$

$$\therefore V_1 = \frac{10,000}{2.303} \times \frac{1}{100} \times (0.041 \text{ liter/atm})$$

(7)

$$V_1 = \underline{\underline{1.78 \text{ liter}}}$$

(b) If $n=2$

$$\therefore W = 2.303 nRT \log \frac{100}{V_1} \\ = 2.303 nRT$$

$$nRT = \frac{W}{2.303}$$

$$RT = \frac{W}{2.303 \times 2}$$

$$T = \frac{10,000}{2.303 \times 2 \times 1.98}$$

(as $n=1.98$)

$$T = \underline{\underline{1100 \text{ K}}}$$

ADIABATIC PROCESSES

In an adiabatic process, there is no exchange of heat between the system and its surroundings. So that $q=0$ and work is done at the expense of the internal energy of the gas. Thus, if work is done on the system, the internal energy of the system will increase. Consequently, if work is done by the system, the internal energy of the system decreases and hence the temperature of the system drops.

Let consider 1 mole of an ideal gas. Since for an adiabatic process, $q=0$, from first law, we have

$$dE = -W = -PdV \quad \text{---} \quad (8)$$

Also from $dE = C_v dT$ (heat capacity at constant volume C_v for an ideal gas, independent of the volume or the pressure of the gas).

$$\text{We have, } C_v dT = -PdV \quad \text{---} \quad (9)$$

for an ideal gas, $P = \frac{N\bar{E}}{V}$, substituting P and rearranging we have,

$$C_v \frac{dT}{T} = -R \frac{dV}{V} \quad \text{---} \quad (10)$$

Here C_v is assumed to be independent of temperature. for a process that takes the gas from a volume V_1 at a temperature T_1 to a new volume V_2 and temperature T_2 . we have,

$$C_v \int_{T_1}^{T_2} \frac{dT}{T} = -R \int_{V_1}^{V_2} \frac{dV}{V}$$

$$\text{or } \frac{C_v}{R} \ln \frac{T_2}{T_1} = -\ln \frac{V_2}{V_1} \quad \text{---} \quad (11)$$

On rearranging equation (11) above, we have,

$$\ln \frac{T_2}{T_1} = \ln \left(\frac{V_1}{V_2} \right)^{R/C_v} \quad \text{---} \quad (12)$$

further, for an ideal gas, we have relations between C_p and C_v as

$$C_p - C_v = R$$

On substituting R in the equation (12), we have

$$\ln \frac{T_2}{T_1} = \ln \left(\frac{V_1}{V_2} \right)^{\frac{\gamma-1}{\gamma}}$$

$$\text{or } \ln \frac{T_2}{T_1} = \ln \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

where $\gamma = \frac{C_p}{C_v}$, the ratio of heat capacities at constant pressure and constant volume. Taking logarithm of both sides of equation (13), we get

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

$$\text{or } T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1}$$

Further, for 1 mole of an ideal gas, $PV = RT$.

$$\text{or } T = \frac{PV}{R}$$

Substituting $T = \frac{PV}{R}$ into equation (14),

$$\frac{P_2 V_2}{R} (V_2)^{\gamma-1} = \frac{P_1 V_1}{R} (V_1)^{\gamma-1}$$

$$\text{or } P_2 V_2^\gamma = P_1 V_1^\gamma \quad \text{--- (15)}$$

Similarly, substituting V by $\frac{RT}{P}$ in equation (15),

$$P_2 \left(\frac{RT_2}{P_2} \right)^\gamma = P_1 \left(\frac{RT_1}{P_1} \right)^\gamma$$

$$\text{or } T_2^\gamma P_2^{1-\gamma} = T_1^\gamma P_1^{1-\gamma} \quad \text{--- (16)}$$

The results of equations (14), (15) and (16) may be summarised in the convenient form

$$TV^{\gamma-1} = \text{constant}, PV^\gamma = \text{constant},$$

$$\text{and } TP^{\frac{1-\gamma}{\gamma}} = \text{constant}$$

(17)

These relationship will ~~not~~ always hold in an adiabatic mechanical change of an ideal gas.

In ~~an~~ isothermal change, $PV = \text{constant}$ and for an adiabatic change $PV^\gamma = \text{constant}$ where $\gamma > 1$.

Work of Expansion in Reversible Adiabatic Process:

from, $W = PdV$.

This equation gives the work of expansion W in an infinitesimal stage of a reversible, adiabatic process, i.e. PdV which is equal to $-C_v dT$ (from equation (9)). For an appreciable process, the total work of expansion, W , is given by,

$$W = \int_{V_1}^{V_2} P dV = - \int_{T_1}^{T_2} C_v dT \quad \text{--- (18)}$$

Assuming $C_v = \text{constant}$, we have on integration,

$$W = -C_v (T_2 - T_1) = C_v (T_1 - T_2).$$

$$\text{or } W = C_v T_1 \left(1 - \frac{T_2}{T_1}\right)$$

$$\text{But since } \frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\therefore W = C_v T_1 \left\{1 - \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}\right\} \quad \text{--- (19)}$$

For Maximum Work in an adiabatic reversible process
of an ideal gas, it follows that,

$$PV^r = \text{constant}$$

On differentiating this equation,

$$rPV^{r-1}dv + V^r dP = 0$$

$$rPdv + VdP = 0$$

$$VdP = -rPdv \quad (20)$$

Also, from complete differential of $PV = RT$, we have

$$Pdv + VdP = R dT$$

$$\text{or } VdP = R dT - Pdv \quad (21)$$

On substituting equation (21) for VdP into equation (20), we get

$$R dT - Pdv = -rPdv$$

$$\text{or } Pdv = \frac{R dT}{(1-r)} \quad (22)$$

$$\text{Further, } W_{\text{max}} = \int_{V_1}^{V_2} Pdv \quad (23)$$

Substituting Pdv from equation (22) into (23) and changing the limits to the temperatures correspondingly to the volumes, we have;

$$W_{\text{max}} = \int_{V_1}^{V_2} Pdv = \int_{T_1}^{T_2} \frac{R dT}{(1-r)}$$

$$\text{or } W_{\text{max}} = \frac{R(T_2 - T_1)}{(1-r)} \quad (24)$$

when $T_2 > T_1$, W_{max} is -ve, meaning work is done on the gas but if $T_2 < T_1$, W_{max} is +ve, work is done by the gas.

Example

Calculate the work of expansion of 1 mole of an ideal gas initially at 25°C, the pressure being changed from 1 to 5 atm under (a) isothermal conditions (b) Adiabatic one (Given $C_p = 7.0 \text{ cal deg}^{-1}$, $C_v = 5.0 \text{ cal deg}^{-1} \text{ mole}^{-1}$).

Solution

(a) Under isothermal conditions.

$$W_{isothermal} = 2.303 RT \log \frac{P_1}{P_2}$$

where, $R = 8.313 \times 10^7 \text{ ergs}$, $T = 298.2 \text{ K}$, $P_1 = 1 \text{ atm}$, $P_2 = 5 \text{ atm}$.

$$\therefore W_{max} = 2.303 \times 8.313 \times 10^7 \times 298.2 \log \frac{1}{5}$$

$$= -3.99 \times 10^{10} \text{ ergs mole}^{-1}$$

(b) Under Adiabatic conditions

$$W = -3.99 \times 10^{10} \text{ ergs mole}^{-1}$$

$$W_{max} = C_v T_1 \left[1 - \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} \right]$$

where $C_v = 5.0 \times 4.187 \times 10^7 \text{ ergs deg}^{-1} \text{ mole}^{-1}$, $T_1 = 298.2 \text{ K}$

$$\gamma = \frac{C_p}{C_v} = \frac{7}{5} \quad \text{or} \quad \frac{\gamma-1}{\gamma} = \frac{2}{7}, \quad P_1 = 1 \text{ atm}, \quad P_2 = 5 \text{ atm}.$$

$$W_{max} = 5.0 \times 4.187 \times 10^7 \times 298.2 \left[1 - \left(\frac{5}{1} \right)^{\frac{2}{7}} \right]$$

$$= -3.649 \times 10^{10} \text{ ergs mole}^{-1}$$

HEAT CONTENT (ENTHALPY, H)

(10)

Most of the chemical processes are performed at constant pressure. In order to deal with processes at this constant, a new thermodynamic function denoted by H will be defined and is called the enthalpy or the heat content of the system. Mathematically;

$$H = E + PV \quad \text{---} \quad \text{---} \quad \text{---} \quad (1)$$

Since E , P and V are all functions of state, therefore H is also a function of state. For a change in enthalpy, ΔH , we get,

$$\Delta H = \Delta E + \Delta(PV) \quad \text{---} \quad \text{---} \quad \text{---} \quad (2)$$

$$\text{or } \Delta H = \Delta E + P\Delta V + V\Delta P \quad \text{---} \quad \text{---} \quad \text{---} \quad (3)$$

$$\text{Since } P = \text{constant } \Delta P = 0$$

$$\therefore \Delta H = \Delta E + P\Delta V \quad \text{---} \quad \text{---} \quad \text{---} \quad (4)$$

$$\text{or } \Delta H = q_P. \quad (\text{as earlier mentioned}) \quad (5)$$

Hence the change in enthalpy of a system is equal to the heat given to the system at constant pressure. The importance of this new state function will become apparent when we study thermochemistry.

HEAT CAPACITIES AT CONSTANT VOLUME AND CONSTANT PRESSURE.

The heat capacity of a system is defined as the heat required to raise the temperature of the system by one degree. If δq is a small quantity of heat added to the system, the temperature of the system

will rise by dT . the heat capacity C of the system is given by:

$$C = \frac{89}{dt} \quad \text{①}$$

for gases, we have two types of heat capacities, at constant volume and at constant pressure.

The heat capacity at constant volume C_v , can be defined as,

$$C_v = \frac{\delta q_{cv}}{dT} \quad \text{---} \quad \text{②}$$

But from first law, we have

$$q = \Delta E + P \Delta V$$

At constant Volume, since $\Delta V = 0$, we have

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$\therefore$ Heat capacity at constant volume is given by

$$\left(\frac{\partial \mathcal{L}}{\partial T} \right)^N$$

the equation (4) above shows that C_v is the rate of change of internal energy with temperature at constant volume.

further when the pressure is held constant, equation takes the form;

$$C_v = \frac{\delta q_T}{\delta T} = \left(\frac{\partial E}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P \quad \text{--- (5)}$$

if $H = E_{\text{top}}$ is differentiated with respect to T at constant pressure, we get,

$$\left(\frac{\partial E}{\partial T}\right)_P = \left(\frac{\partial E}{\partial T}\right)_P + P \left(\frac{\partial V}{\partial T}\right)_P \quad \text{--- (6) (16)}$$

Consequently, $C_P = \left(\frac{\partial H}{\partial T}\right)_P$ --- (7)

the difference between the two heat capacities specific is given by;

$$C_P - C_V = \left(\frac{\partial H}{\partial T}\right)_P - \left(\frac{\partial E}{\partial T}\right)_V \quad \text{--- (8)}$$

On substituting equation (5) for $\left(\frac{\partial H}{\partial T}\right)_P$ into equation (8) we have, i.e. at constant P

$$C_P - C_V = \left(\frac{\partial E}{\partial T}\right)_P + P \left(\frac{\partial V}{\partial T}\right)_P - \left(\frac{\partial E}{\partial T}\right)_V \quad \text{--- (9)}$$

Since the internal energy E is in general a function of any 2 of 3 variables, P, V, T , we take T, V as independent variables i.e. $E = f(T, V)$.

$$dE = \left(\frac{\partial E}{\partial T}\right)_V dT + \left(\frac{\partial E}{\partial V}\right)_T dV \quad \text{--- (10)}$$

Dividing equation (10) by dT at a constant pressure,

$$\left(\frac{\partial E}{\partial T}\right)_P = \left(\frac{\partial E}{\partial T}\right)_V + \left(\frac{\partial E}{\partial V}\right)_T \quad \text{--- (11)}$$

Putting equation (11) into equation (9), we have

$$C_P - C_V = \left(\frac{\partial E}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P + \left(\frac{\partial E}{\partial T}\right)_V + P \left(\frac{\partial V}{\partial T}\right)_P$$

$$\text{or } C_P - C_V = \left[\left(\frac{\partial E}{\partial V}\right)_T + P\right] \left(\frac{\partial V}{\partial T}\right)_P \quad \text{--- (12)}$$

In the above equation (12), $P \left(\frac{\partial V}{\partial T} \right)_P$ is the work done $P dV$ per unit increase in temperature in a constant pressure process. The other term, $\left(\frac{\partial E}{\partial V} \right)_T$, $\left(\frac{\partial V}{\partial T} \right)_P$ is the contribution from the energy required for the change in Volume to pull the molecules farther apart against the cohesive or repulsive intermolecular forces. ~~the~~

For ideal gas, $\left(\frac{\partial E}{\partial V} \right)_T = 0$ i.e. the internal energy is independent of Volume at constant temperature, i.e. the energy of the gas is a function of temperature only. Hence equation (12) becomes,

$$C_p - C_v = P \left(\frac{\partial V}{\partial T} \right)_P \quad \text{--- (13)}$$

further, for one mole of an ideal gas $PV = RT$, then $P \left(\frac{\partial V}{\partial T} \right)_P = R$

$$\text{Hence, } C_p - C_v = R \quad \text{--- (14)}$$

In other words, the heat capacities at constant pressure and constant volume differ from each other by a constant amount equal to the gas constant $R = 1.987 \text{ cal/deg-mole}$.

Examples

① For a gas which obeys the equation of state $PV = RT$, Use the equation $\left(\frac{\partial E}{\partial V} \right)_T = T \left(\frac{\partial P}{\partial T} \right)_V - P$ to show that $\left(\frac{\partial E}{\partial V} \right)_T = 0$.

~~② For a gas~~

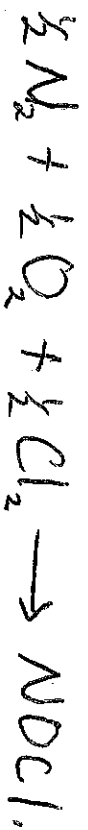
② Derive a mathematical expression which consists of only constants and easily measurable physical

Parameters for work done, W by a gas in the surrounding medium. The gas obey an equation of state $PV = nRT - \frac{n^2 a}{V}$ and expands reversibly from V_1 to V_2 at constant T .

(3) The heat change in enthalpy for the formation of NDCl from the gaseous element is $12.57 \text{ kcal mol}^{-1}$ at 25°C . If the gases are ideal, calculate the change in internal energy ΔE . ($1 \text{ kcal} = 4.184 \text{ kJ}$).

Solution:

(3) Chemical equation



$$\frac{1}{2} \text{ mole} = 1 \text{ mole}$$

$\therefore$ 1 mole of product - 1 $\frac{1}{2}$ mole of reactant = $-\frac{1}{2}$ mole = n .
from 1st law of thermodynamics

$$\Delta H = \Delta E + \Delta W$$

$$= \Delta E + P \Delta V$$

$$= \Delta E + nRT$$

$$\therefore \Delta E = \Delta H - nRT$$

but $\Delta H = 12.57 \text{ kcal mol}^{-1}$, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, $T = 298 \text{ K}$, $n = \frac{1}{2} \text{ mole}$.

$$\therefore \Delta E = (12.57 \text{ kcal mol}^{-1}) - \left(-\frac{1}{2}\right)(8.314 \text{ J mol}^{-1} \text{ K}^{-1})(298 \text{ K})$$

$$= 12.51.36 \text{ kcal mol}^{-1}$$

$$\text{or } \Delta E = 12.51.36 \text{ kcal mol}^{-1} \times (4.184 \text{ kJ})$$

$$\Delta E = 5235.69 \text{ kJ}$$

$$\therefore \text{The change in internal energy } \Delta E = \underline{\underline{5235.69 \text{ kJ}}}$$

Solution ①

$$\text{Hence } PV = RT \quad \dots \text{--- (1)}$$

$$\left(\frac{\partial E}{\partial V}\right)_T = T \left(\frac{\partial P}{\partial T}\right)_V - P \quad \dots \text{--- (2)}$$

$$\text{To show } \left(\frac{\partial E}{\partial V}\right)_T = 0.$$

Taking partial derivative by part 2 equation (1).

$$P \partial V + V \partial P = R \partial T \quad \dots \text{--- (3)}$$

$$\text{but at constant } V, \partial V = 0.$$

$$V \partial P = R \partial T \quad \dots \text{--- (4)}$$

Rearrange equation (4) at constant V

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V} \quad \dots \text{--- (5)}$$

Multiply both sides by T .

$$T \left(\frac{\partial P}{\partial T}\right)_V = \frac{RT}{V} \quad \dots \text{--- (6)}$$

Substitute $P = \frac{RT}{V}$ into equation (6).

$$T \left(\frac{\partial P}{\partial T}\right)_V = P. \quad \dots \text{--- (7)}$$

Now, putting (7) into (2) above, we have.

$$\left(\frac{\partial E}{\partial V}\right)_T = T \left(\frac{\partial P}{\partial T}\right)_V - T \left(\frac{\partial P}{\partial T}\right)_V$$

$$\Rightarrow \left(\frac{\partial E}{\partial V}\right)_T = 0. \quad \text{QED.}$$

Solution: ②

$$PV = nRT - \frac{n^2 a}{V} \quad \text{--- (1)}$$

$$P = \frac{nRT}{V} - \frac{n^2 a}{V^2} \quad \text{--- (2)}$$

$$\text{but } dW = P dV$$

$$\therefore dW = \left(\frac{nRT}{V} - \frac{n^2 a}{V^2} \right) dV \quad \text{--- (3)}$$

by integrating equation (3)

$$W = nRT \int_{V_1}^{V_2} \frac{1}{V} dV - n^2 a \int_{V_1}^{V_2} \frac{1}{V^2} dV \quad \text{--- (4)}$$

$$= nRT \ln \frac{V_2}{V_1} - \left[\frac{n^2 a}{V^2} + \frac{n^2 a}{V_1} \right]$$

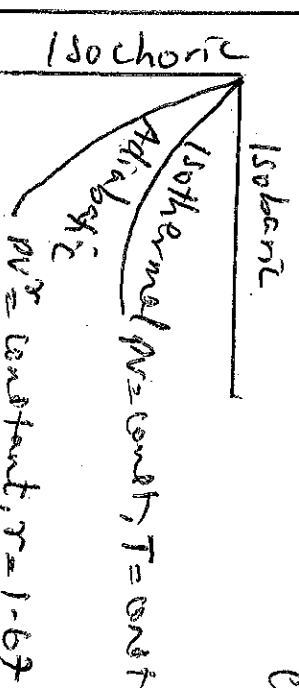
$$= nRT \ln \frac{V_2}{V_1} - \frac{n^2 a}{V_2} + \frac{n^2 a}{V_1}$$

$$= nRT \ln \frac{V_2}{V_1} - n^2 a \left(\frac{1}{V_2} + \frac{1}{V_1} \right) \quad \text{Ans.}$$

Isobaric Processes.

Those processes which take place at constant pressure are called isobaric processes. For example, heating water to its boiling point and its vaporisation take place at the same atmospheric pressure. For isobaric process $dp = 0$.

Isobaric process at
constant volume, $dV = 0$.



Four basic thermodynamic processes.